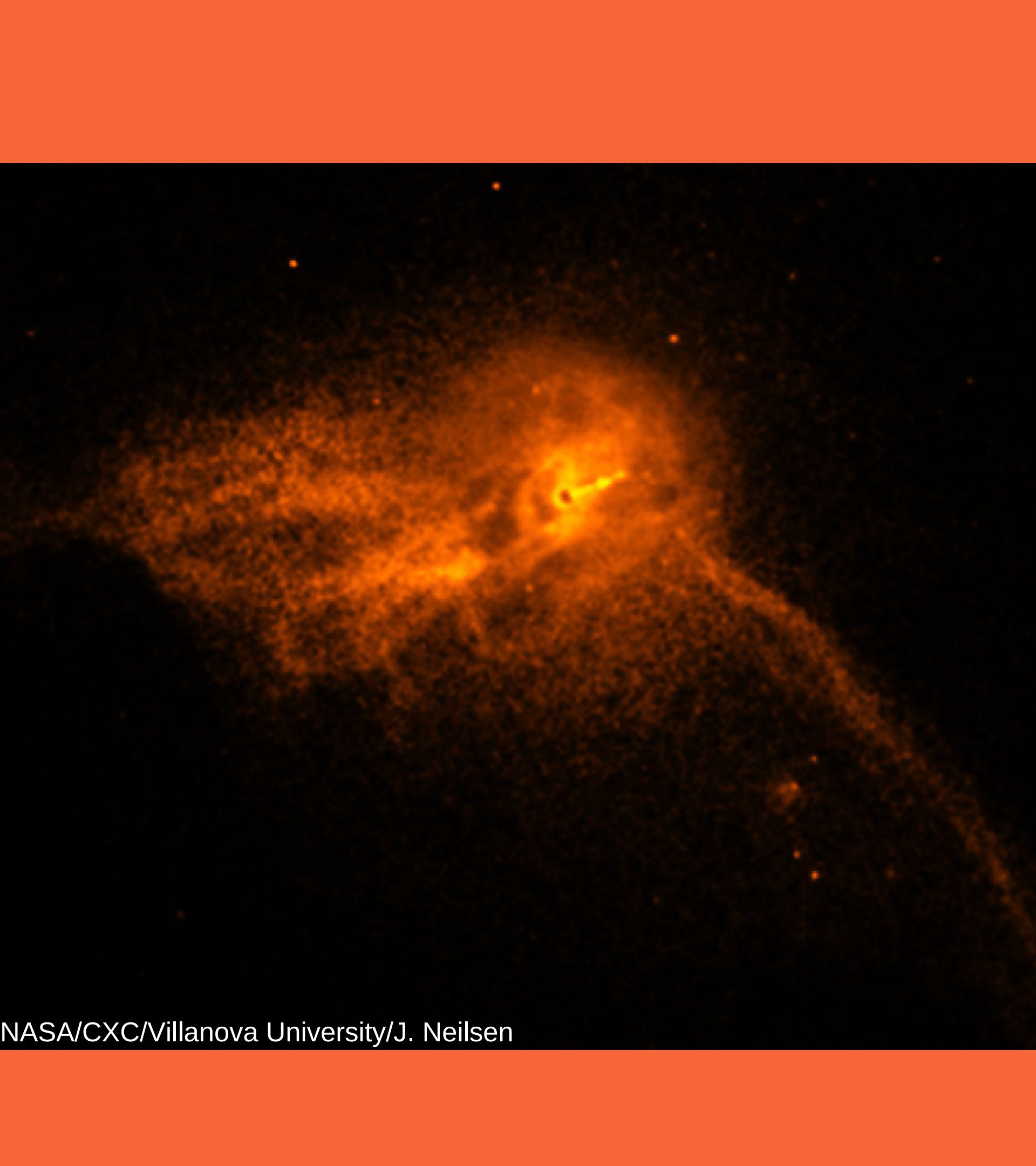


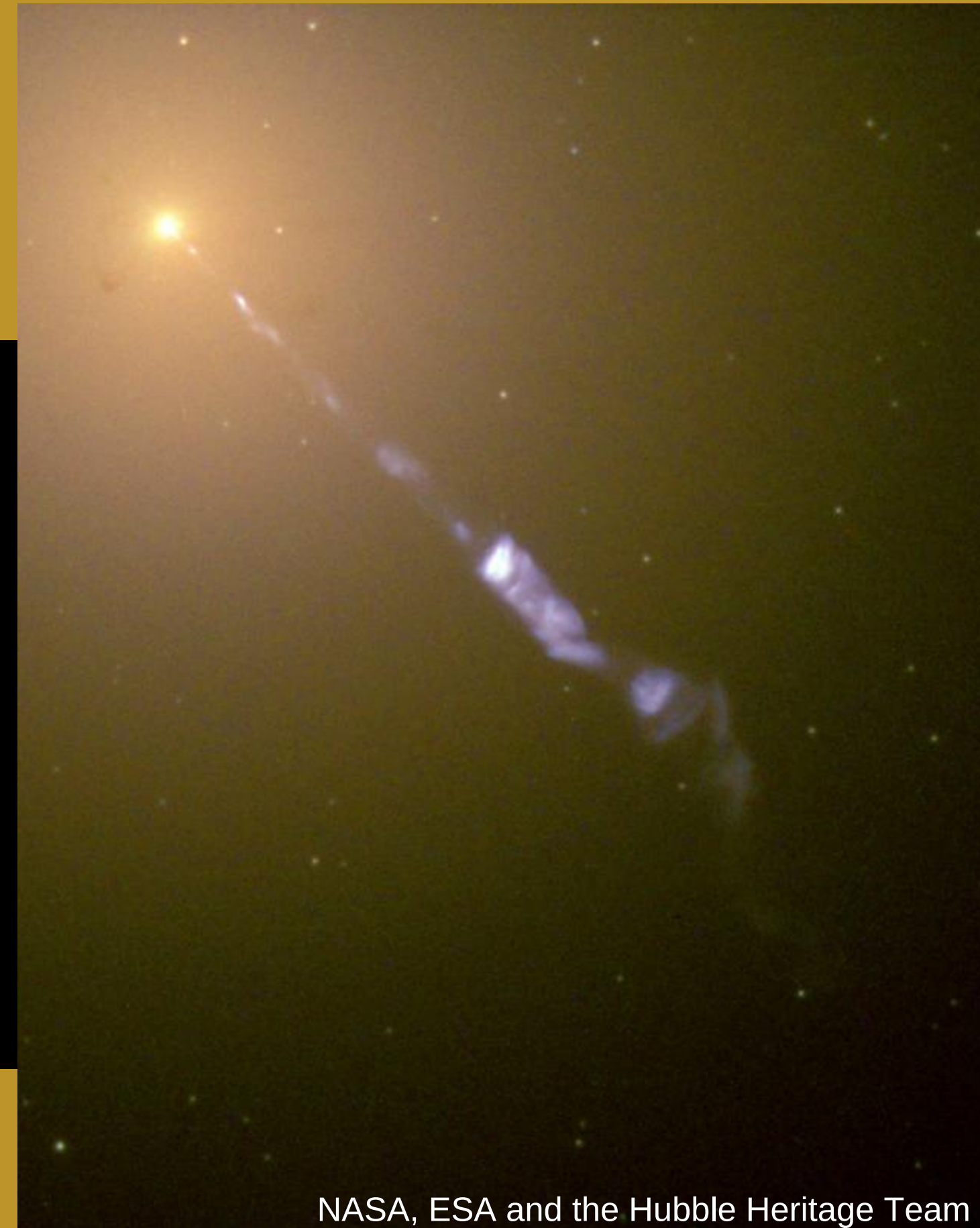
A large, diffuse, orange-red X-ray emission region, likely the hot gas in the core of the M87 galaxy, is shown against a black background. The emission is concentrated in a central, bright, irregularly shaped area with some filamentary structure. Several small, distinct orange-red spots are visible in the surrounding black space, representing other X-ray sources.

X-ray Variability and the Spectrum of M87

C. Sheridan & J. Neilsen

Outline

- **Why M87? Why X-rays?**
- **Spectra and SEDs**
- **Power-Laws and Breaks**
- **Main Objectives**
- **The Dataset**
- **Spectral Modeling**
- **Flux Variability**
- **Conclusions**



NASA, ESA and the Hubble Heritage Team

Why M87?

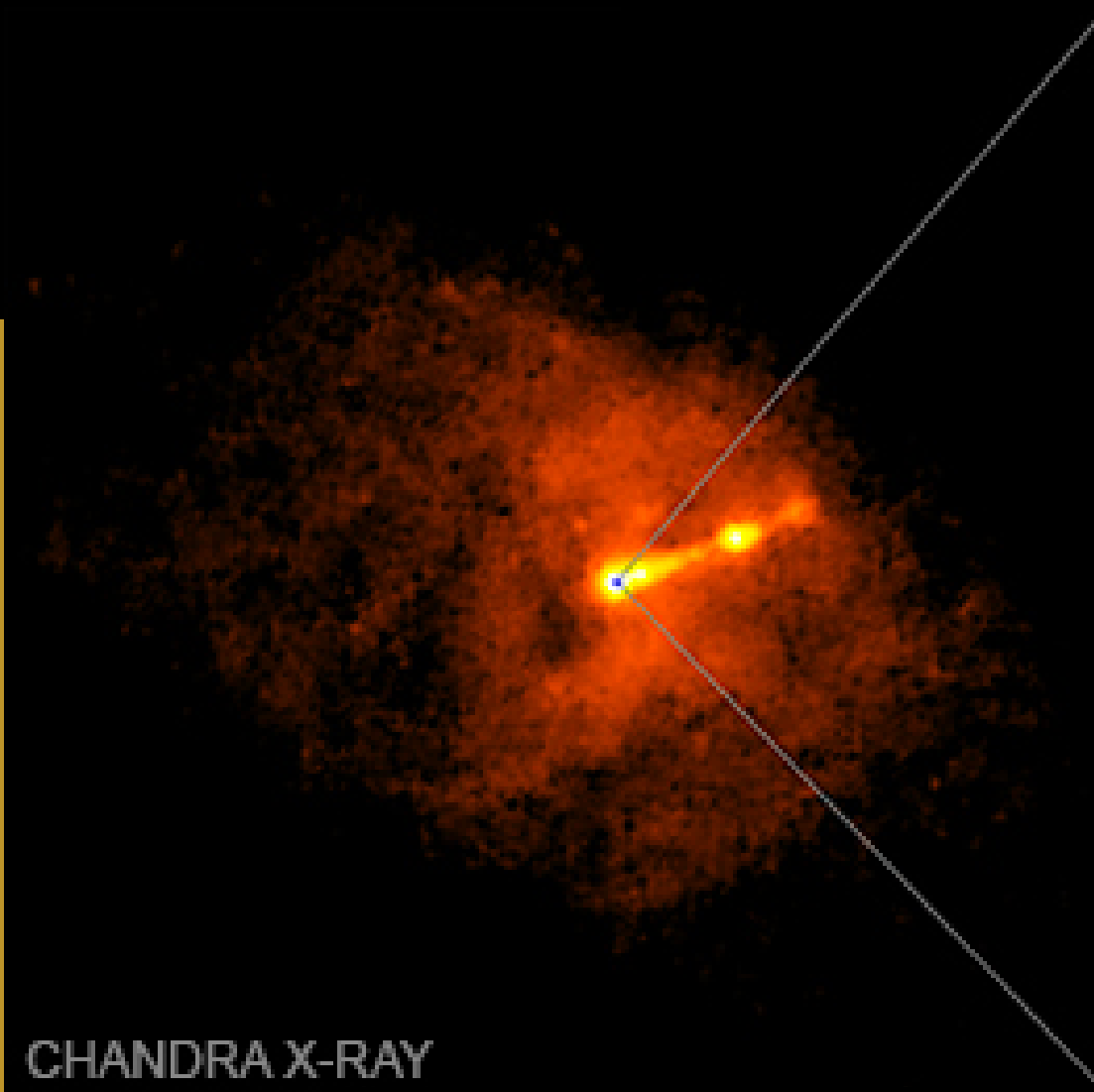
What is M87?

- Large elliptical galaxy within the Virgo Cluster, home to a several billion $M_{\odot}$ SMBH
- Has a large relativistic jet that it ejects into intergalactic space
- First black hole (shadow) to ever be directly imaged!!

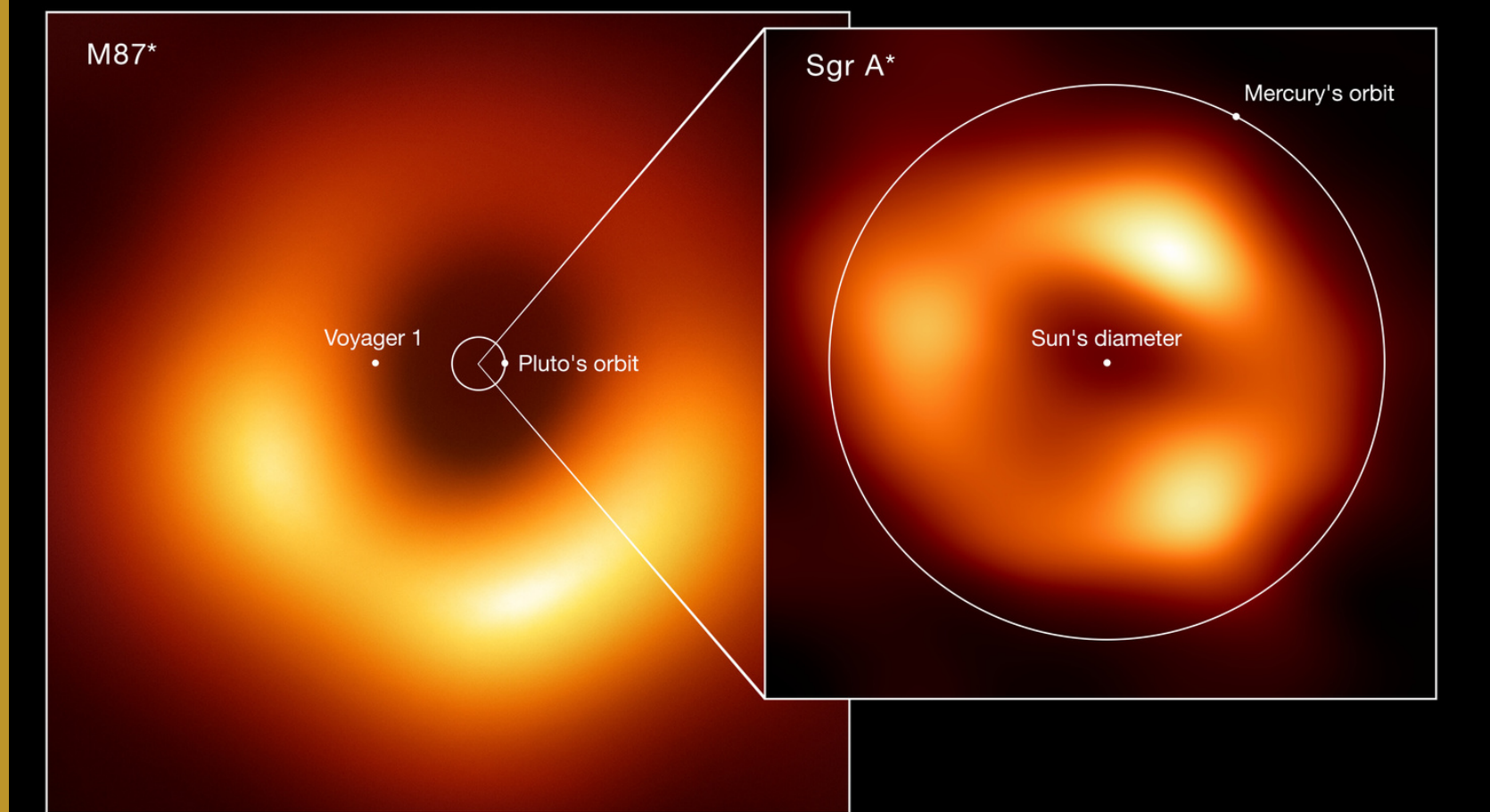
Why M87?

- Not very distant, very large, very bright source
- Key target for the Event Horizon Telescope

CHANDRA X-RAY



EHT



M87 Structure

X-Ray

Virgo Cluster



CHANDRA X-RAY

M87 Nucleus July 17, 2002
HST STIS/MAMA

Optical/UV

Core



Jet

HST-1

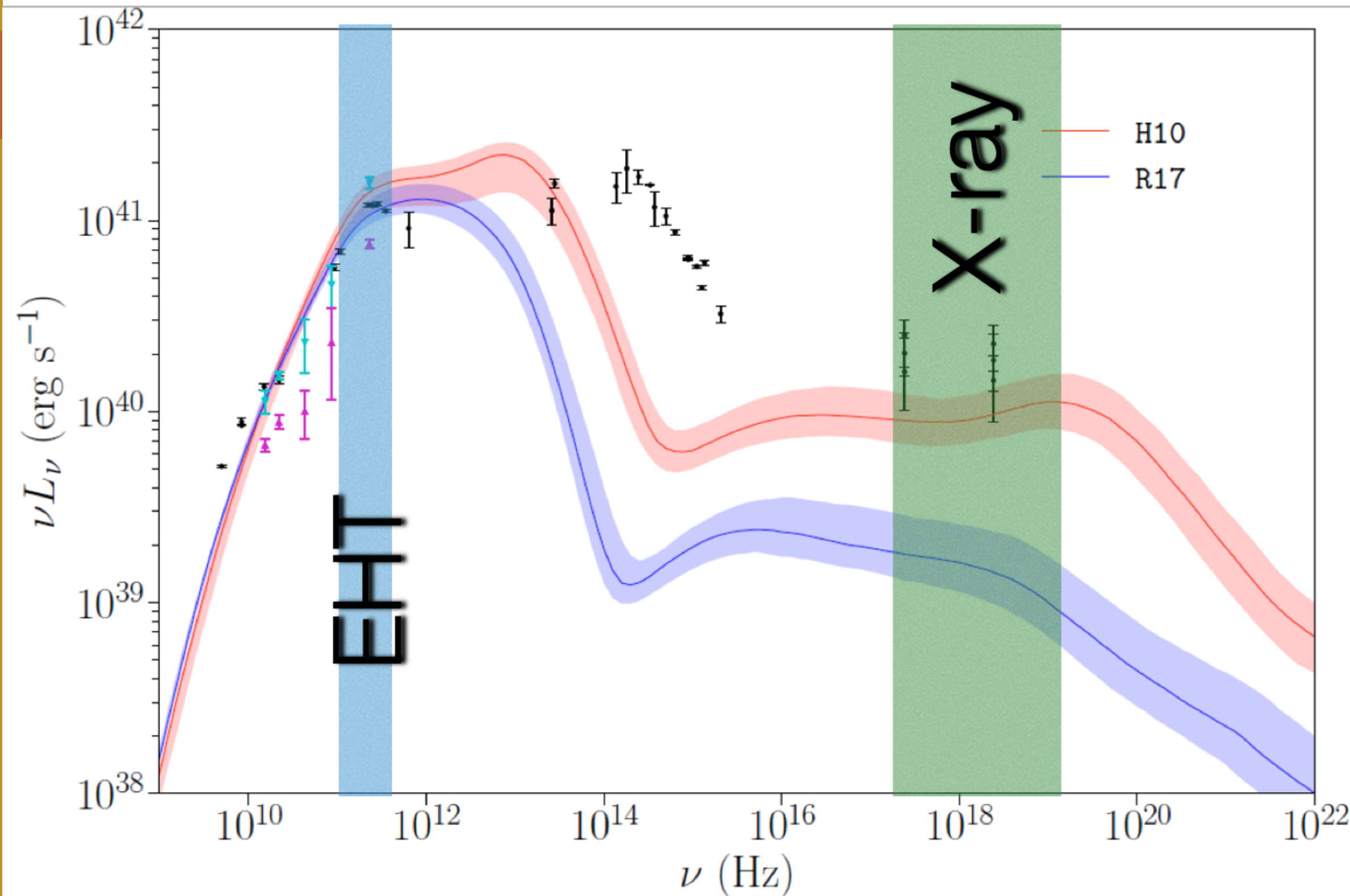


500 light-years
153 pc 1".9

Focusing on X-rays!

Who cares about X-rays?

- We do! We are particularly interested in particle acceleration in jets = "microphysics"
- X-rays are very sensitive to things that the Event Horizon Telescope (EHT) cannot see, like microphysics!



Spectra & SEDs

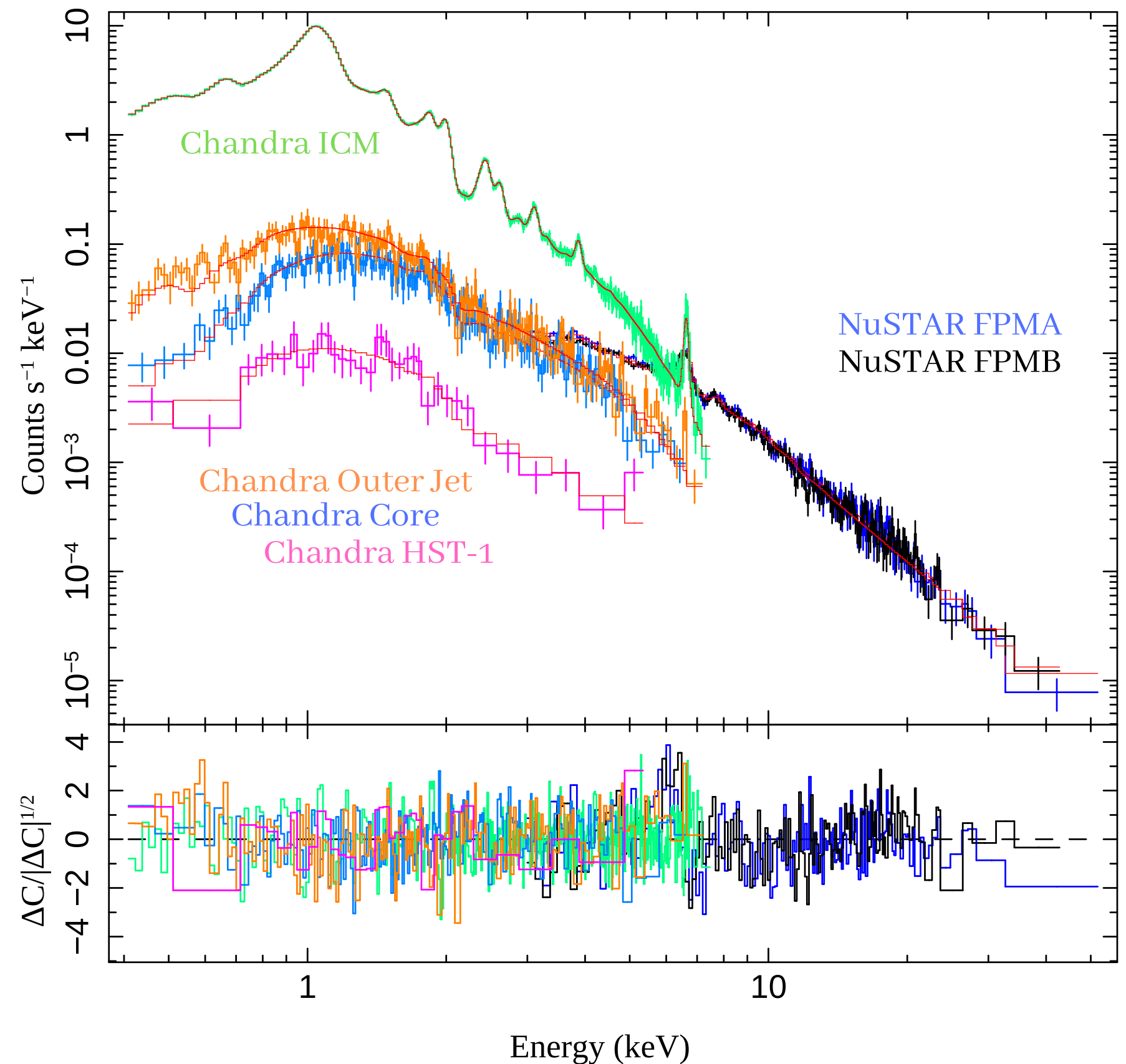
What is a spectrum?

- A spectrum is a plot of intensity vs. energy level or frequency.
- Provide lots of insight about a system's characteristics and behavior!

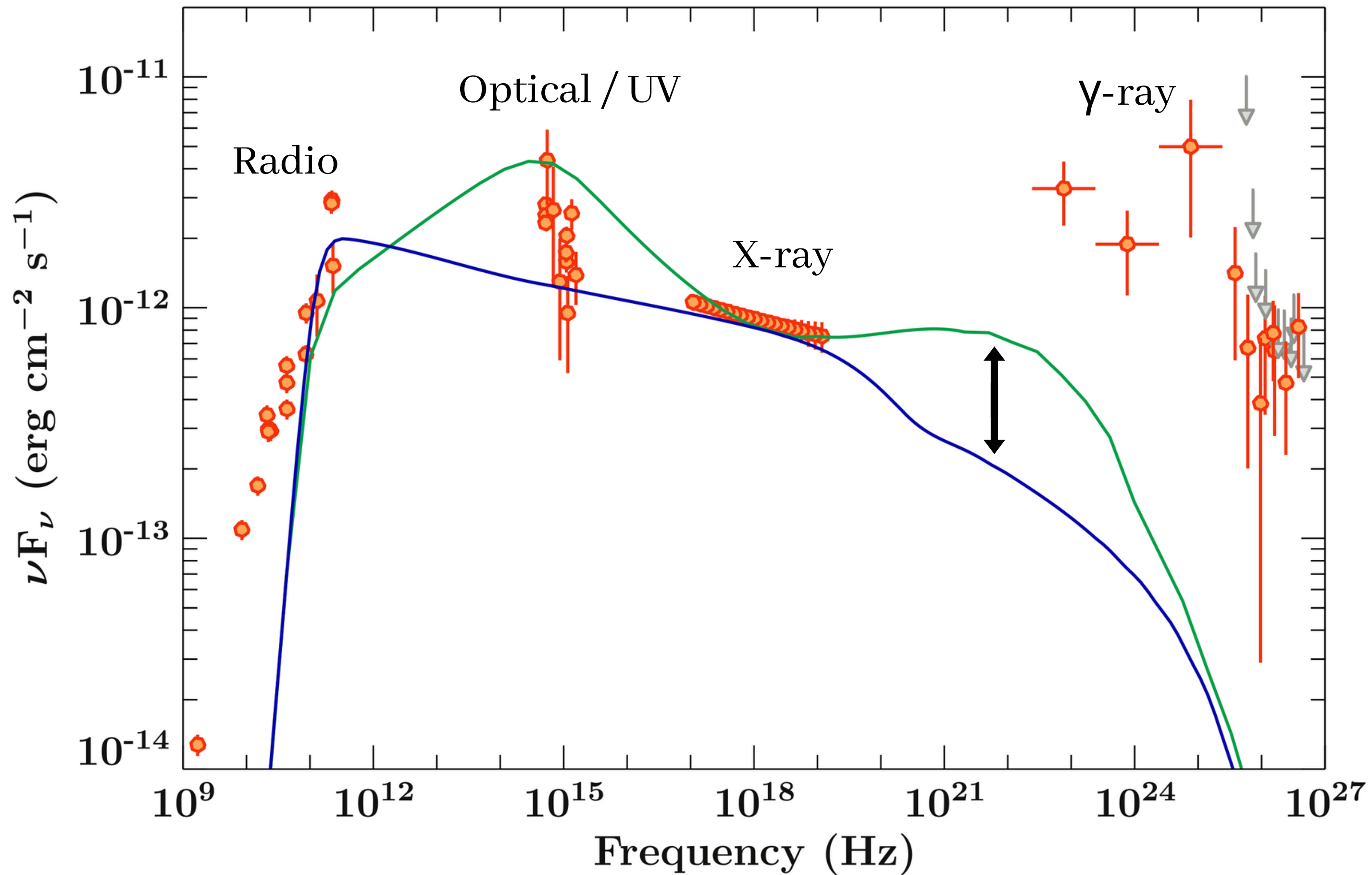
What is an SED?

- Spectral Energy Distribution
- Similar to a spectrum, but encompass entire MUCH broader energy ranges; from radio out to γ -rays!

Stacked M87 Spectrum w/ Broken Power-Law Fit

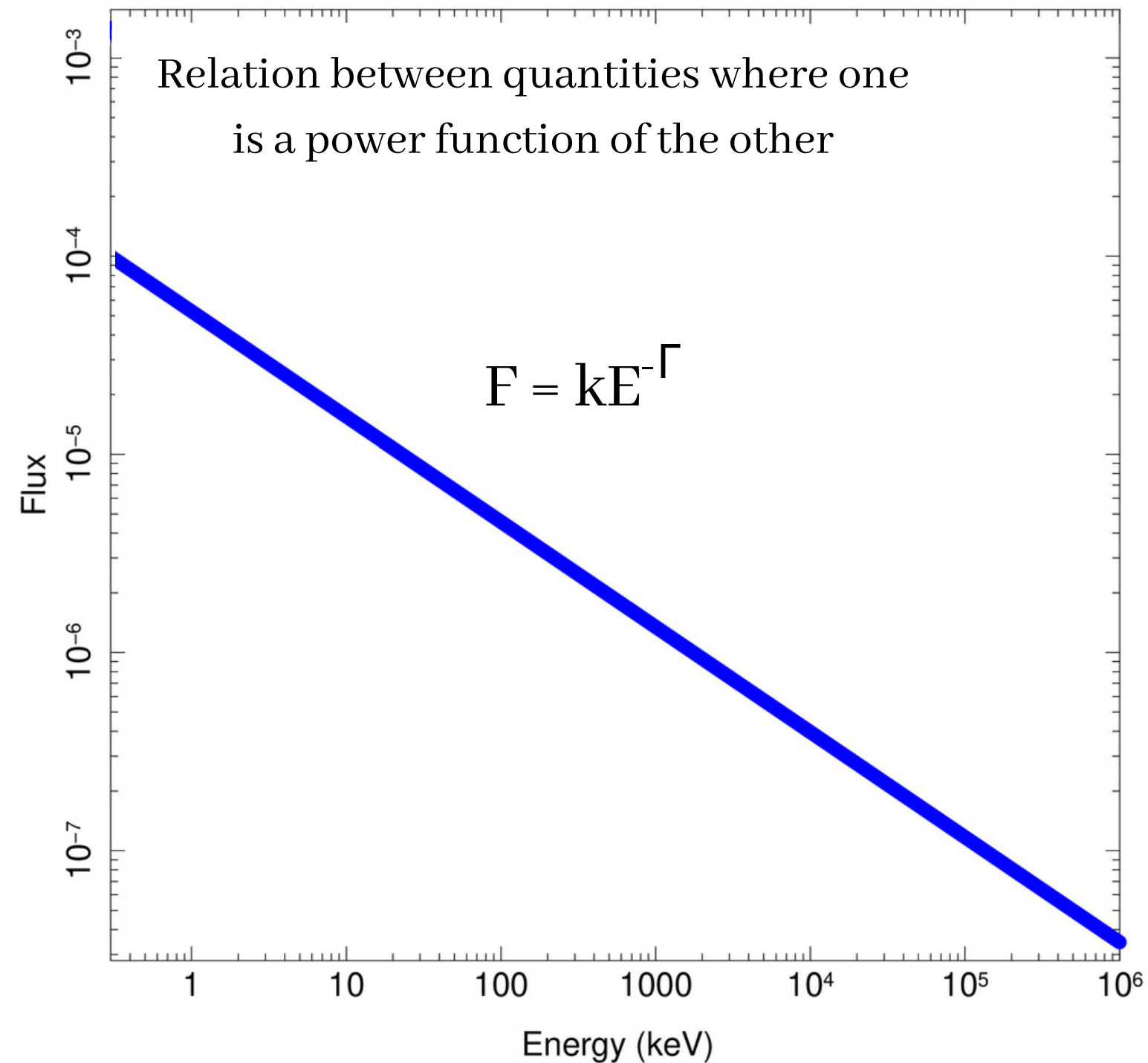


M87 Spectral Energy Distribution (SED)

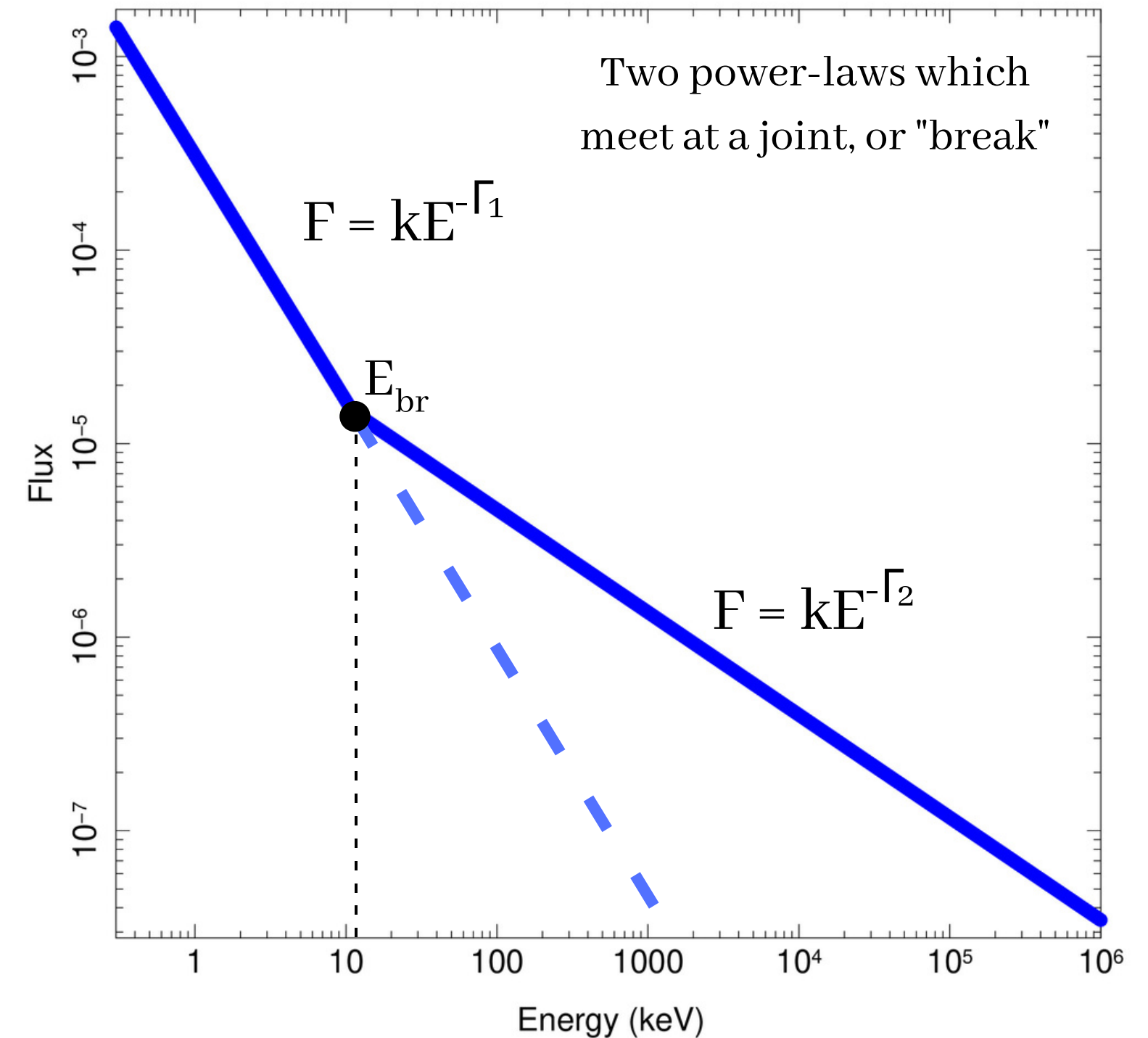


Power-Law & Broken Power-Laws

Power-Law Example



Broken Power-Law Example



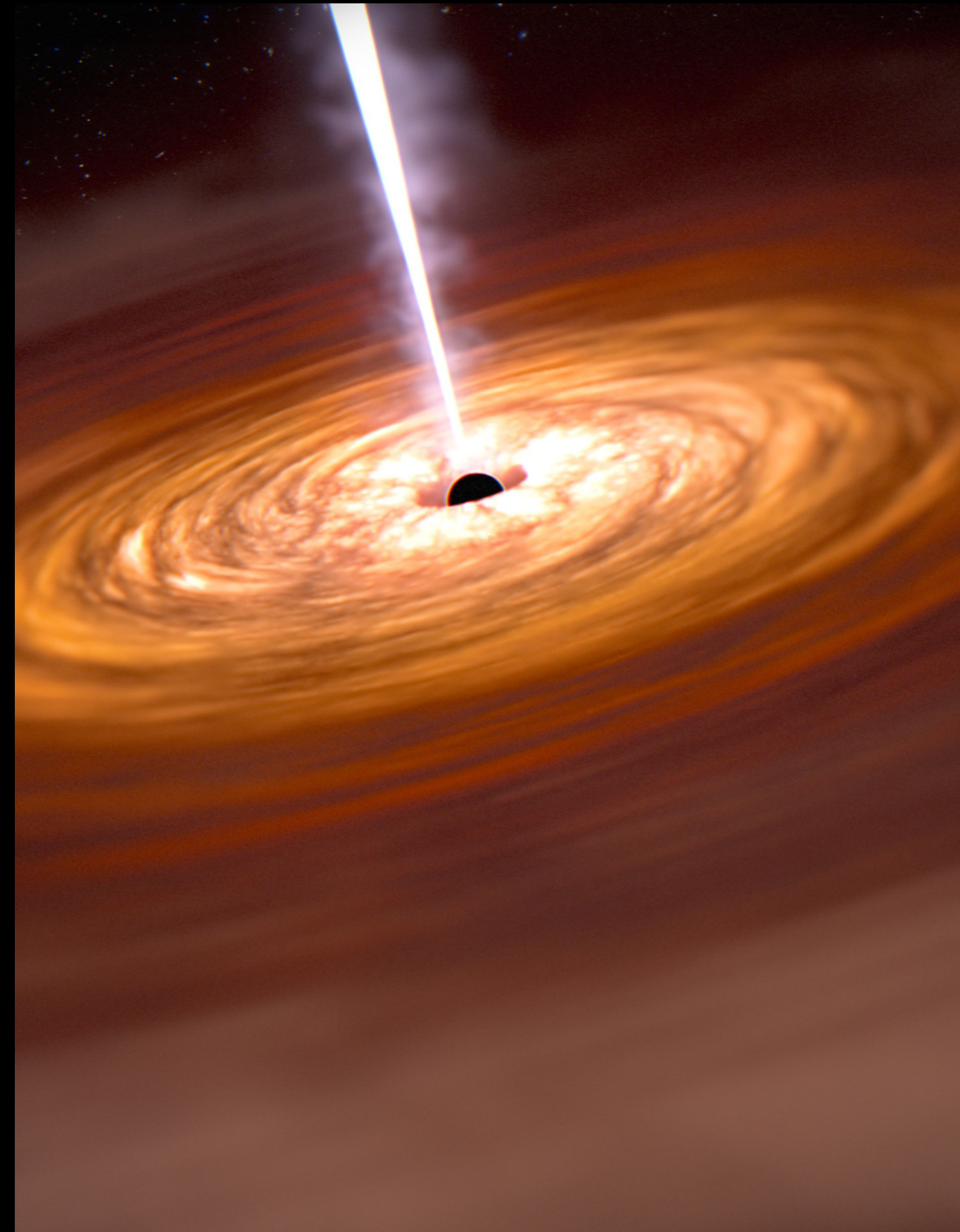
Our Main Objectives

1. Determine the core spectrum.

Under the assumption that the spectral shape of M87 is consistent, we will determine the spectral shape of both the overall system and the core.

2. Observe core flux variability.

With the spectral shape constrained, we can use that consistent shape to observe the X-ray flux variability of the core over our observing timescale (2017 April – 2022 June)



The Dataset

NuSTAR Observations (top left)

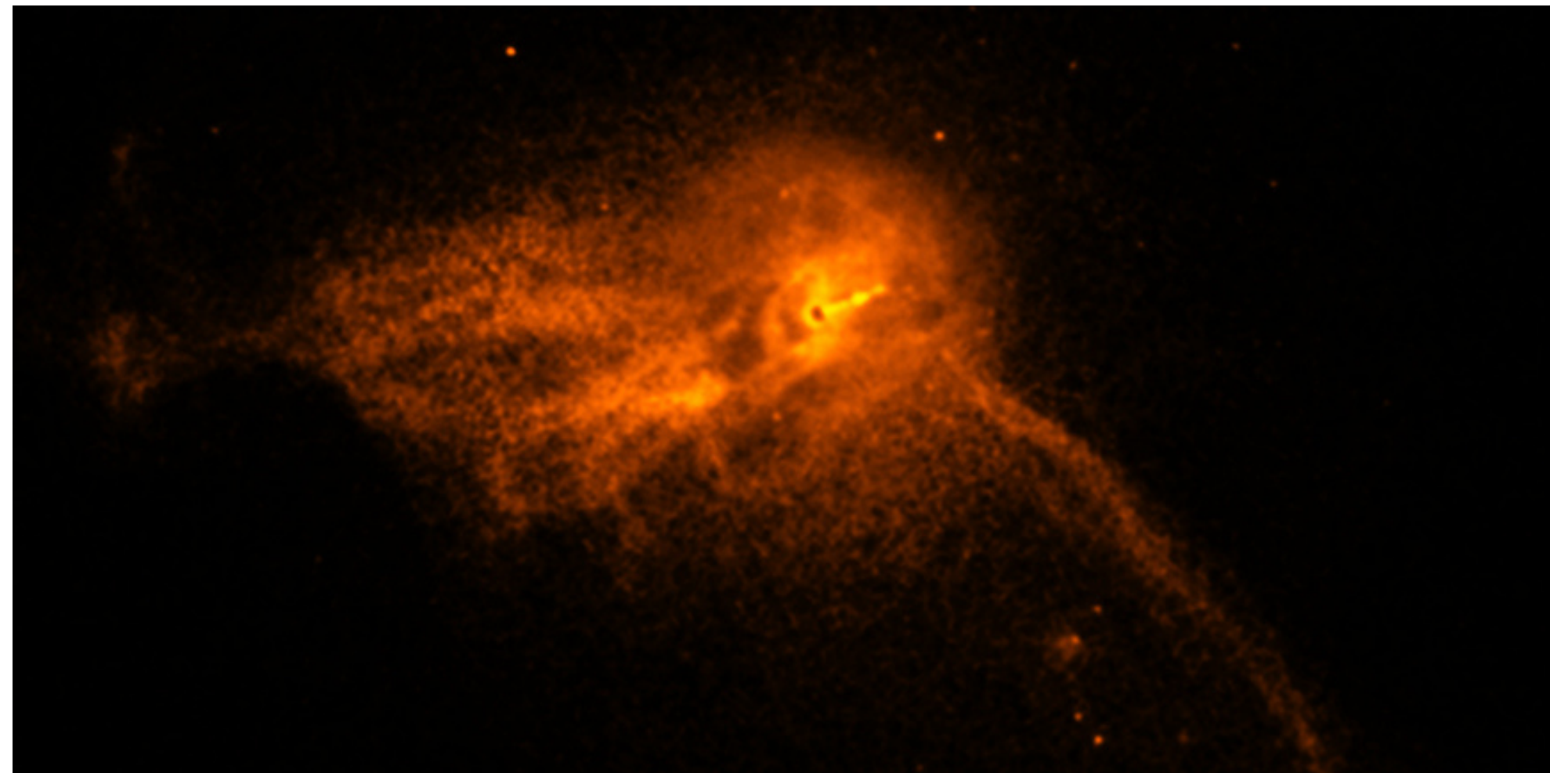
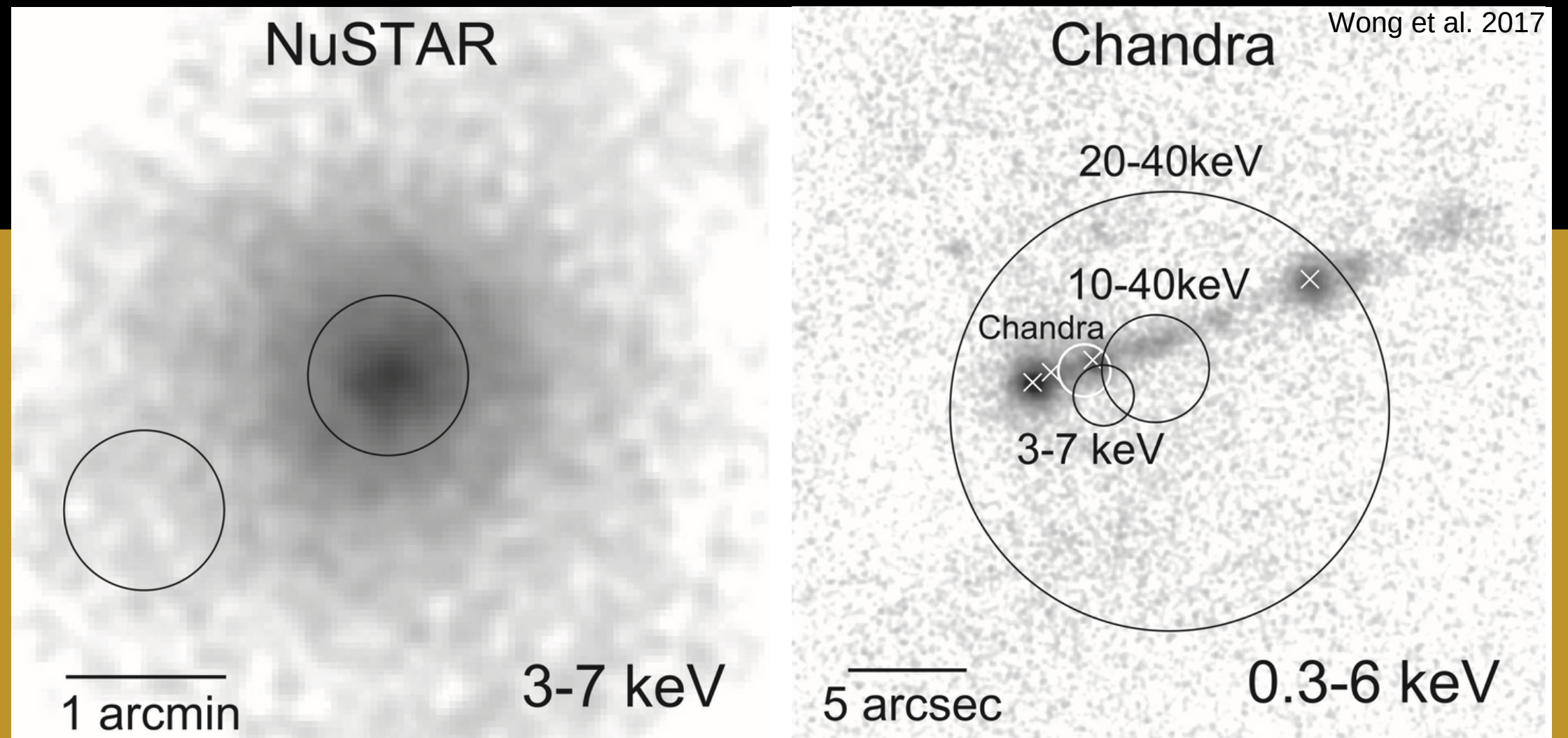
14 *NuSTAR* observations between 2017 April and 2022 June of M87

Chandra Jet Components (top right)

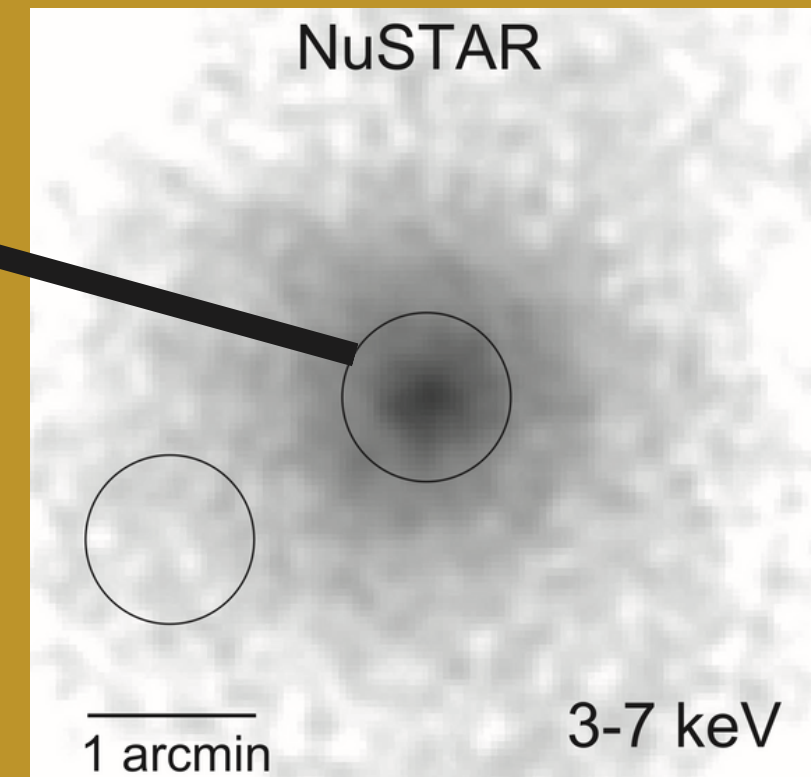
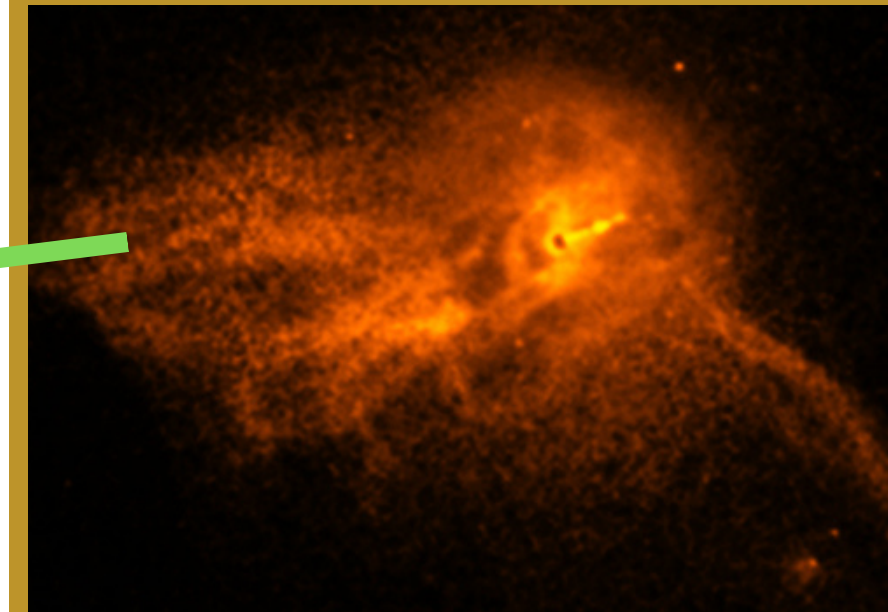
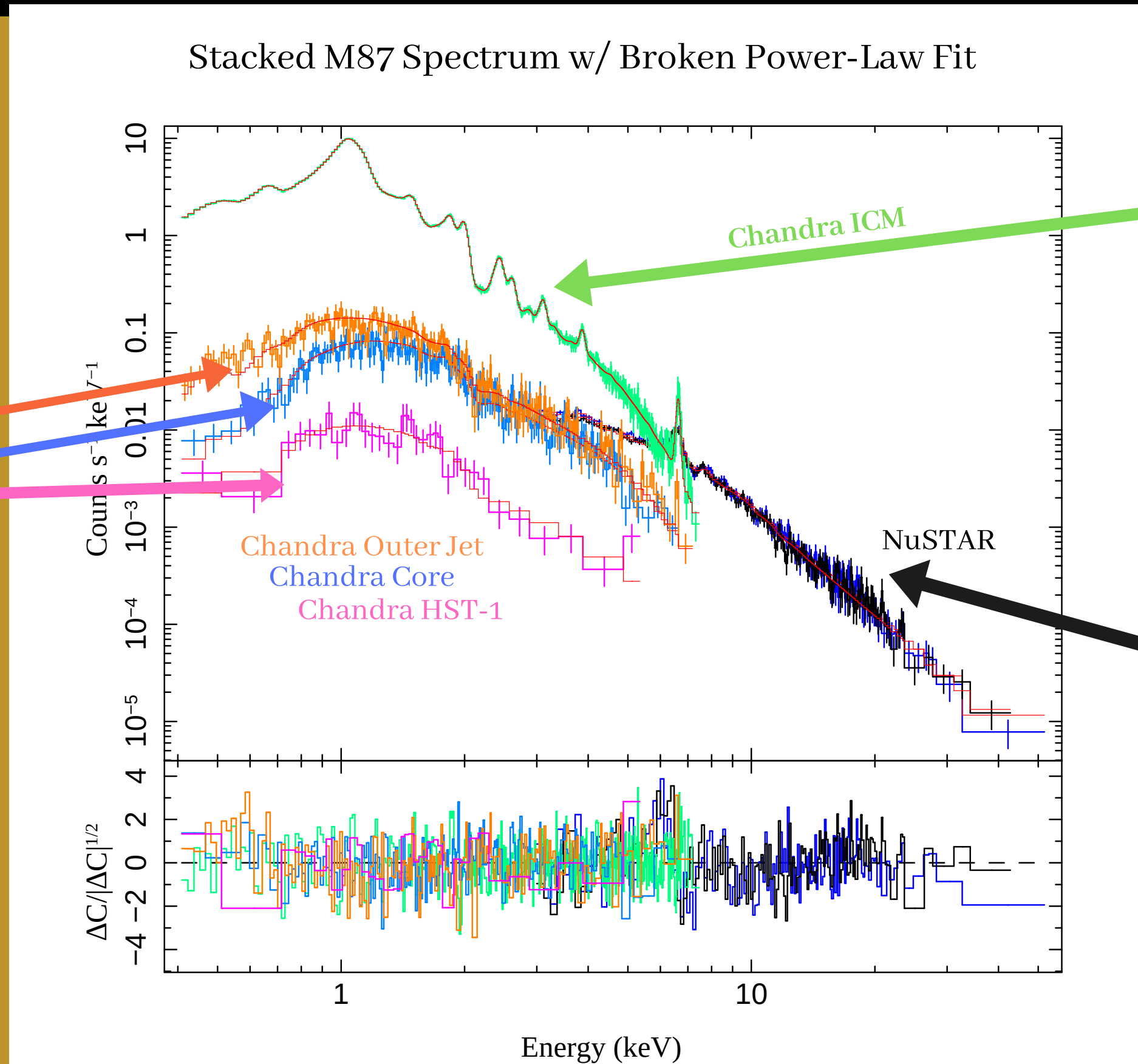
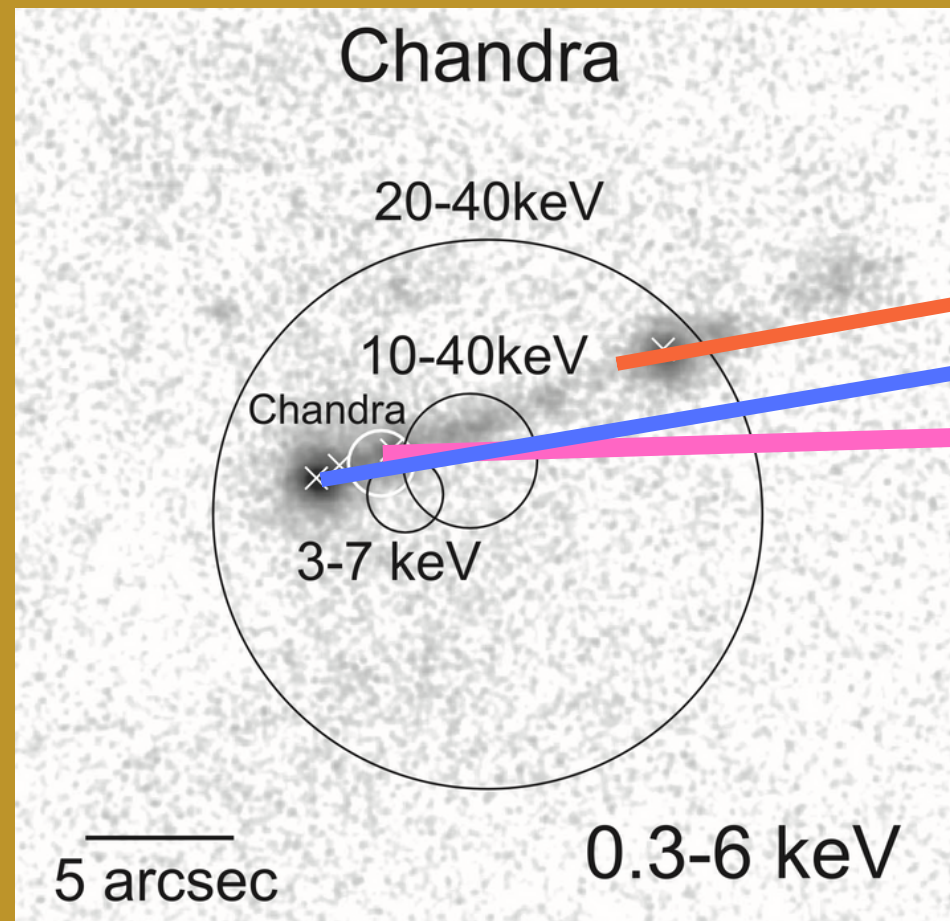
Chandra spatially resolved observations of the core, HST-1, and the outer jet

Chandra Virgo Cluster (bottom)

3 *Chandra* observations from July 2000 and July 2002 of the surrounding Virgo Cluster

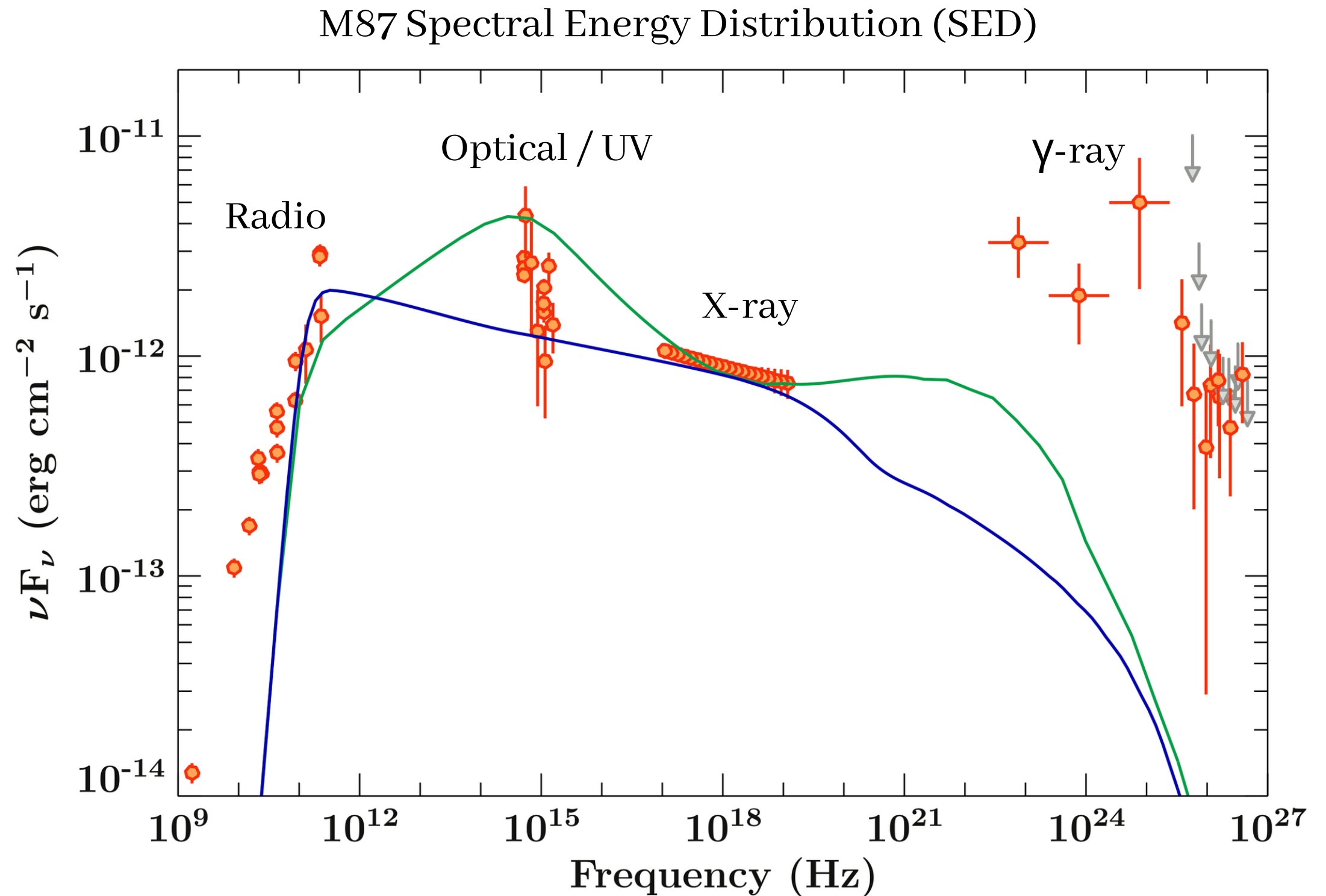


Components in the Spectra



Why a Broken Power-Law?

- Historically, X-ray spectra of M87 have been fit using a simple power-law...
- Models w/ a power-law (blue) under-predict γ -ray emission...
- Some SED models (green) call for a break in the X-ray!



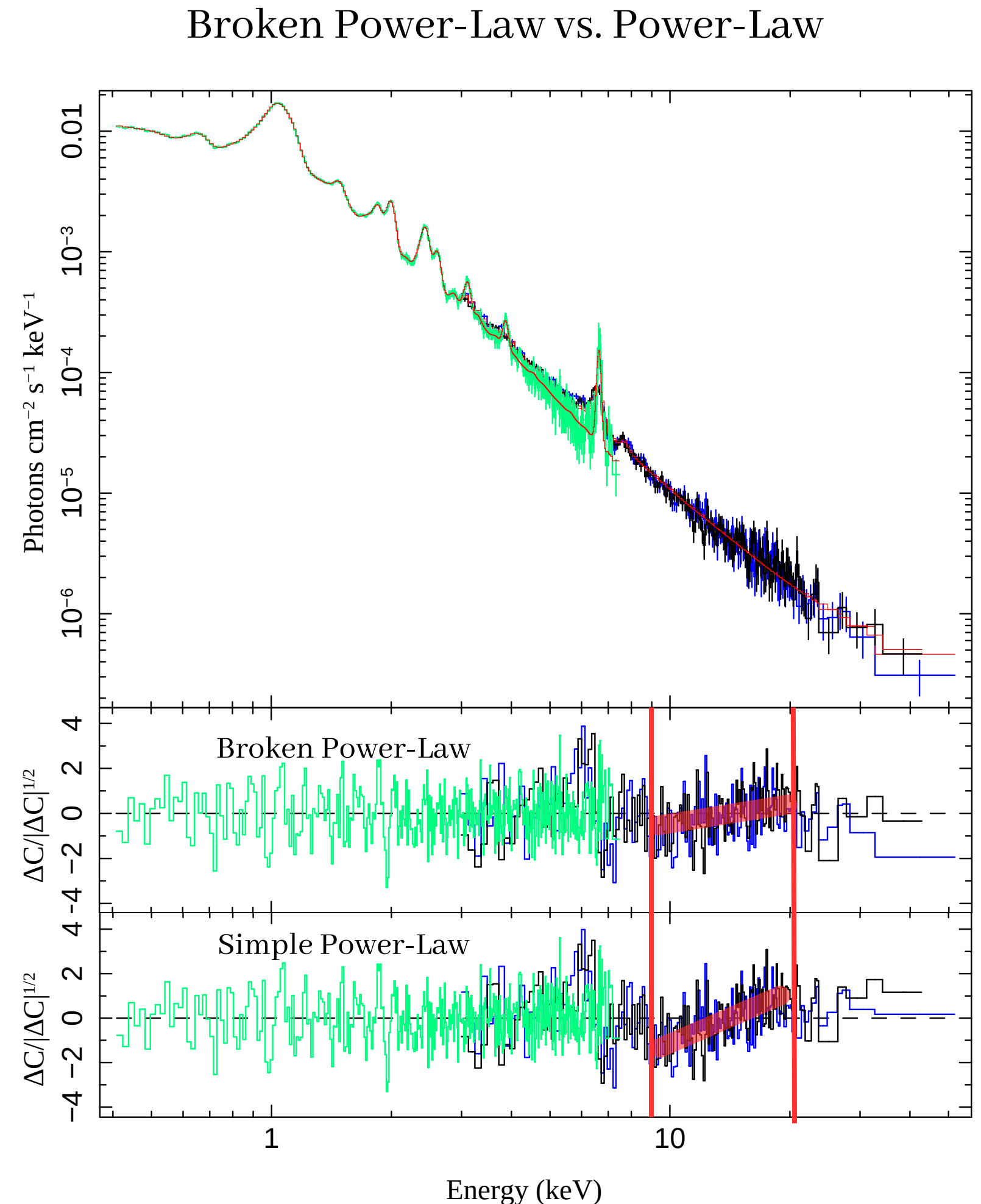
Broken Power-Law Results

Quality of Fit

Broken Power-law saw a statistically significant improvement over the simple power-law

Broken Power-Law

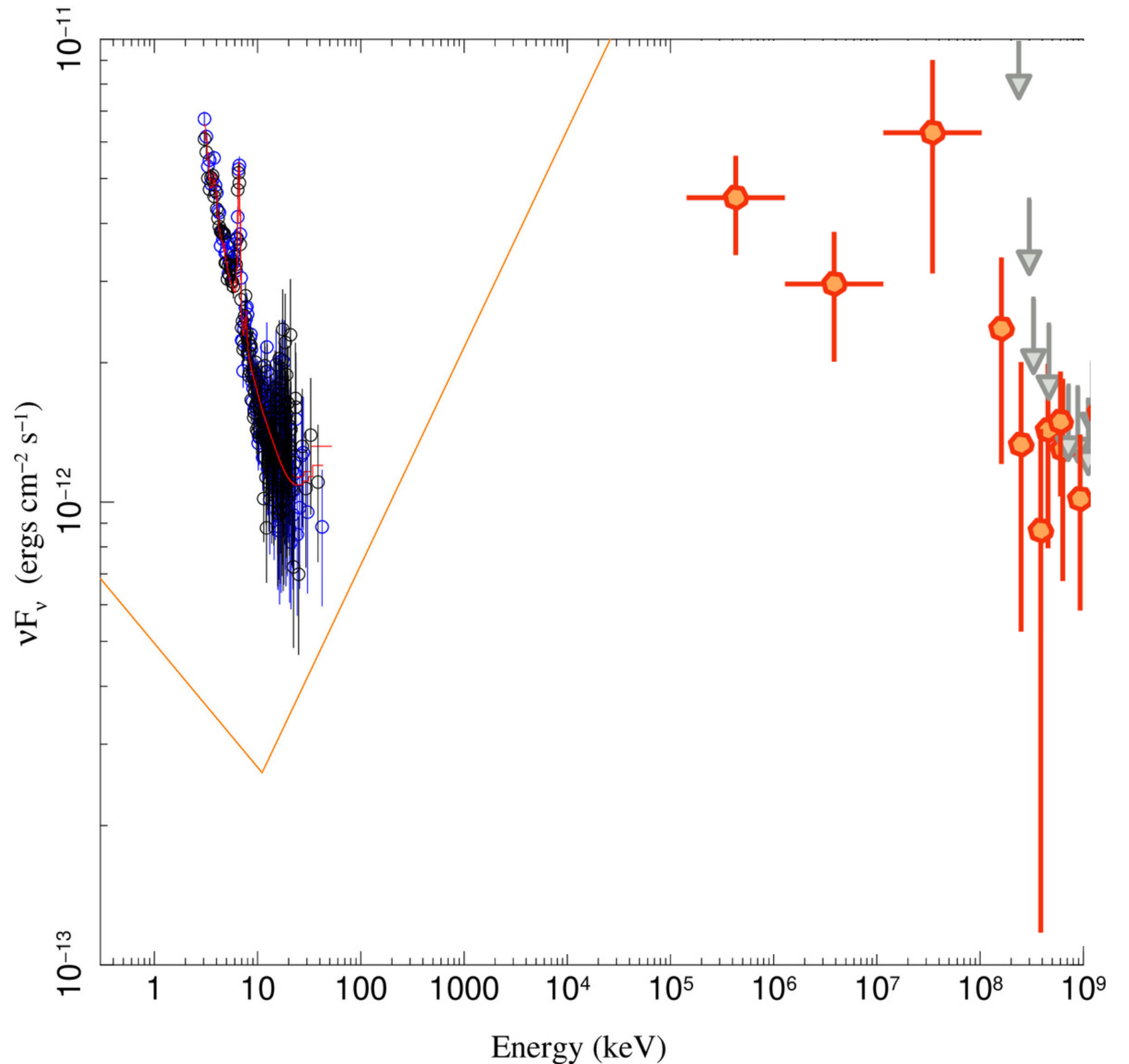
$\Gamma 1$	2.26	+0.06 / -0.05
E _{br}	11 keV	+/- 1 keV
$\Gamma 2$	1.6	+0.1 / -0.2



Extrapolated Broken Power-Law

What does this tell us?

- The broken power-law dramatically overestimates γ -ray emission, but...
- A broken power-law extrapolated out to TeV level energies is not physically realistic, but this opens the door for modeling with an X-ray break!



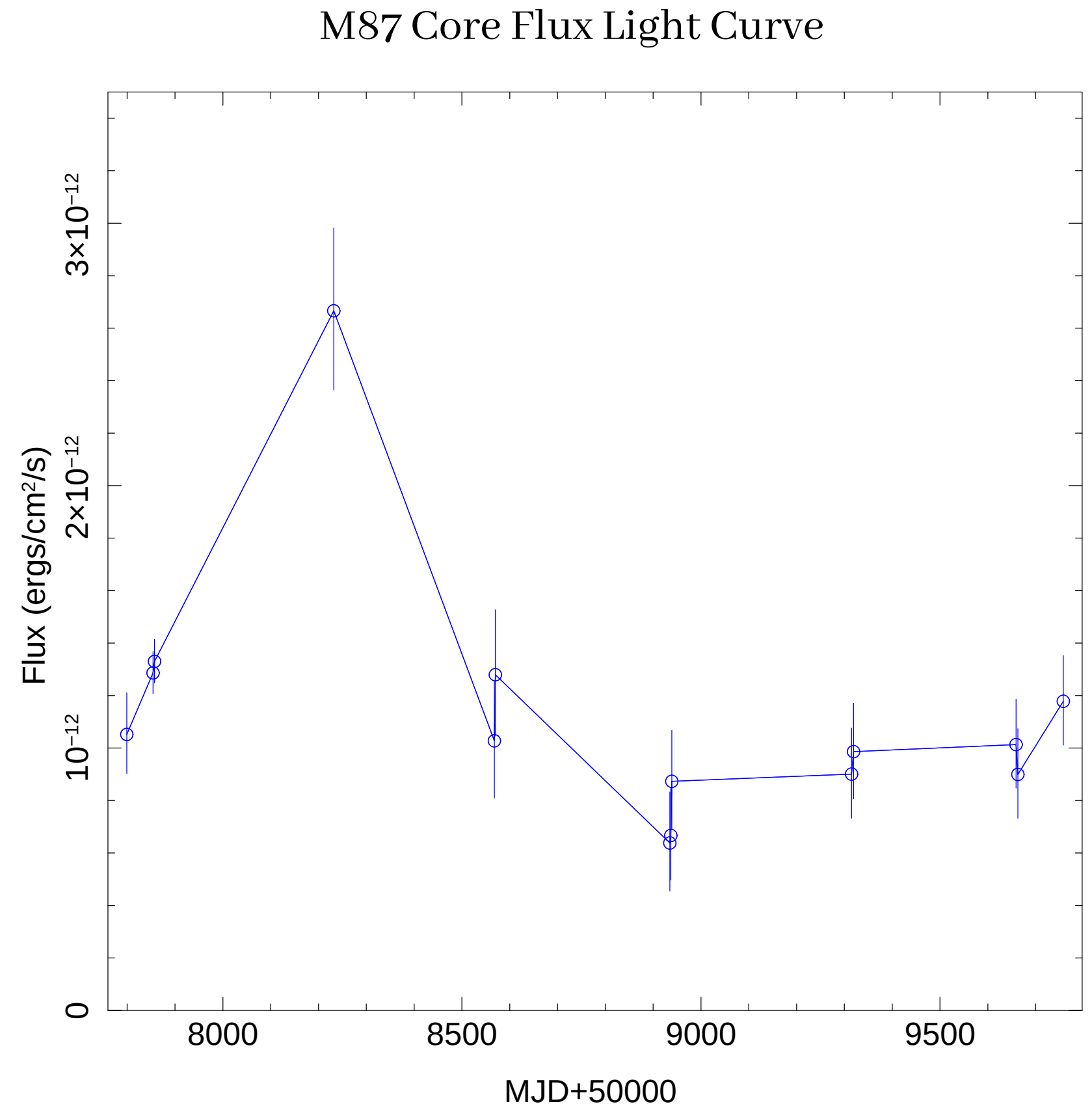
The Light Curve

Core Flux Variability

Over our observing timescale, the flux of M87 is relatively consistent, with the exception of a flared state in 2018.

Flared State (2018)

We investigated the 2018 spectrum for any noticeable spectral differences compared to other observations, but found no significant differences



Conclusion

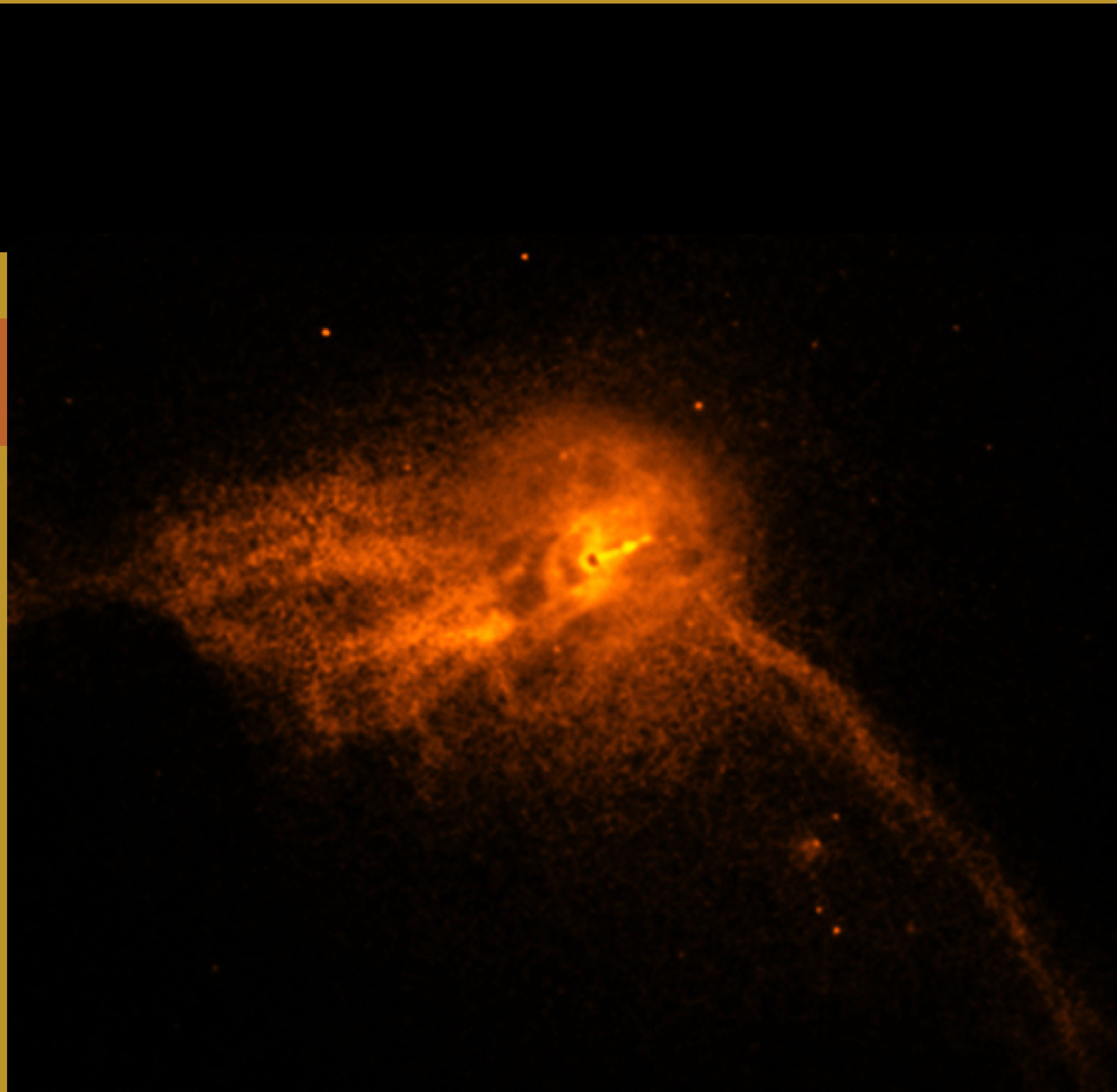
4 Main Points of Conclusion:

1. **M87 appears to have a consistent spectral shape!**
2. **No significant variability in the core flux!**
3. **Can confirm the presence of a break in the spectra!**
4. **That break helps to account for the γ -ray emission!**

Future Study

Major Point of Focus:

We want to incorporate the broken power-law model into ongoing EHTC analysis of M87!



Questions?